

Income and Life Expectancy across Countries

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Draft

Abstract

People in richer countries live longer. Using data on 142 countries observed every five years from 1952 to 2007, we show that average life expectancy rose from 49.1 to 67.0 years. In 2007, a 10 percent higher GDP per capita is associated with 0.73 more years of life expectancy.

1 Introduction

Preston (1975) showed that life expectancy rises with income across countries, steeply among poor countries and slowly among rich ones. We revisit this relationship with more recent data.

2 Data

We use the Gapminder data distributed by Bryan (2025). They cover 142 countries every five years from 1952 to 2007. Life expectancy is measured in years at birth. GDP per capita is measured in inflation-adjusted US dollars.

3 Results

Average life expectancy rose from 49.1 years in 1952 to 67.0 years in 2007. In 2007 it ranged from 39.6 years in Swaziland to 82.6 years in Japan, a gap of 42.8 years.

Table 1 reports averages by continent for 2007. Africa, with 14.9 percent of the population in the sample, had the lowest average life expectancy, 54.8 years, and the lowest median GDP per capita, \$1,452.

Figure 1 plots life expectancy against GDP per capita on a log scale. The correlation between life expectancy and the log of GDP per capita is 0.81. A regression of life expectancy on the log of GDP per capita gives a slope of 7.2, so a 10 percent higher GDP per capita is associated with 0.73 more years of life expectancy.

References

Bryan, J. (2025). gapminder: Data from Gapminder. R package. <https://doi.org/10.32614/CRAN.package.gapminder>

Preston, S. H. (1975). The changing relation between mortality and level of economic development. *Population Studies* 29(2), 231–248. <https://doi.org/10.1080/00324728.1975.10410201>

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Table 1: Countries, life expectancy and income by continent, 2007

Continent	Countries	Life expectancy (years)	Median GDP per capita (US dollars)
Africa	52	54.8	1,452
Americas	25	73.6	8,948
Asia	33	69.2	4,471
Europe	30	77.6	28,504
Oceania	2	80.7	29,810
All	142	67.0	6,124

Notes: Life expectancy is the unweighted average across countries. Source: Gapminder.

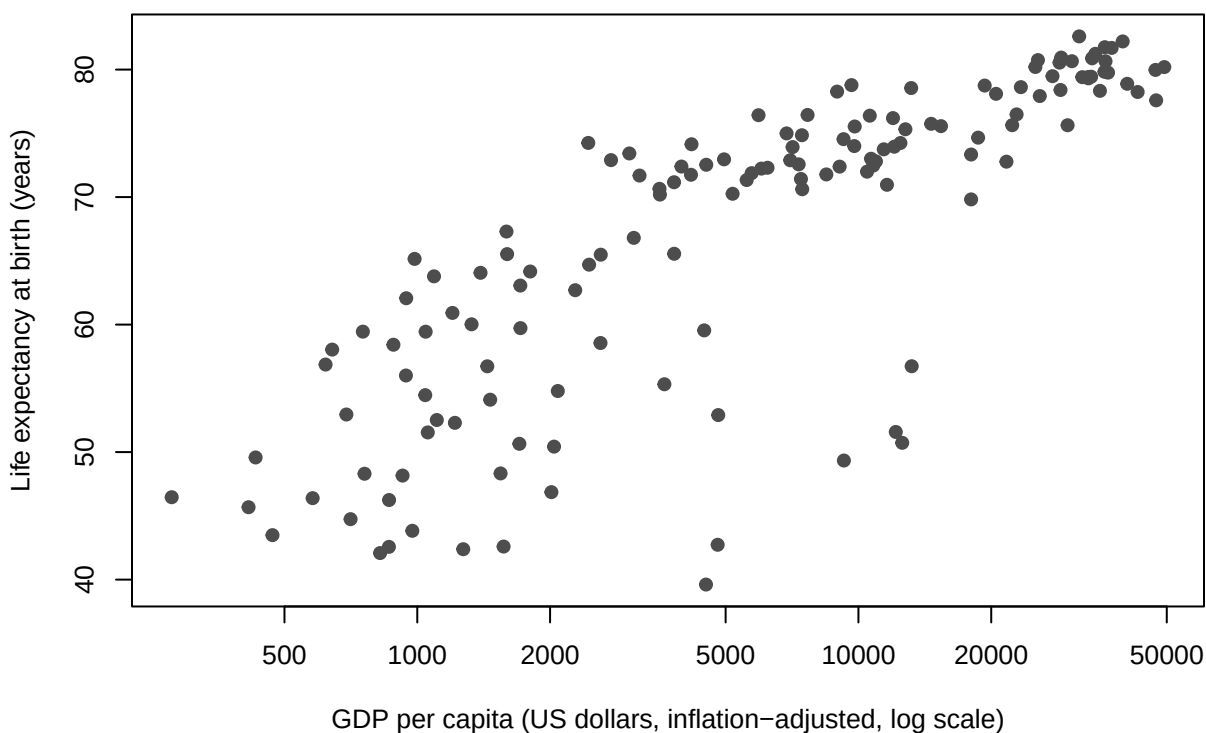


Figure 1: Life expectancy and GDP per capita, 2007

Notes: Each dot is a country. The horizontal axis is on a log scale. Source: Gapminder, 2002.